Python Programming Assignment

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GitHub Repository

https://github.com/username/repository

Assignment No: 1

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1 Variables

What is it?

1. A variable is a name used to store a value.
2. The value can be changed during the program.

Why do we need it?

1. To store data for later use.
2. To make programs easy to write and understand.

How is it Used?

1. Used to store numbers, text, and other data.
2. Created using the `=` operator.

2 Basic Data Types

2.1 Integer

What is it?

1. An integer is a whole number.
2. Example: `10`, `-5`, `100`.

Why do we need it?

1. To perform counting and calculations.
2. To store values without decimals.

How is it Used?

1. Used for age, marks, quantity, etc.
2. Example: `age = 20`

2.2 Float

What is it?

1. A float is a number with decimal values.
2. Example: `3.14`, `25.5`.

Why do we need it?

1. To store decimal numbers.
2. Useful for measurements and calculations.

How is it Used?

1. Used for height, weight, price, etc.
2. Example: `price = 99.99`

2.3 Boolean

What is it?

1. A Boolean has only two values: True or False.
2. It represents yes/no conditions.

Why do we need it?

1. To make decisions in programs.
2. Used in conditions and loops.

How is it Used?

1. Used with if statements.
2. Example: `isPassed = True`

2.4 String

What is it?

1. A string is a collection of characters.
2. It is written inside quotes.

Why do we need it?

1. To store text data.
2. Useful for names, messages, and addresses.

How is it Used?

1. Used to display or process text.
2. Example: `name = "Alice"`

3 Arithmetic Operators

What is it?

1. Arithmetic operators perform mathematical calculations.
2. Examples: `+`, `-`, `*`, `/`.

Why do we need it?

1. To calculate values.
2. To solve mathematical problems.

How is it Used?

1. Used in formulas and calculations.
2. Example: `total = a + b`

4 Comparison Operators

What is it?

1. Comparison operators compare two values.
2. They return True or False.

Why do we need it?

1. To check conditions.
2. Used for decision making.

How is it Used?

1. Used in if statements.
2. Example: `a > b`

5 Assignment Operators

What is it?

1. Assignment operators assign values to variables.
2. Example: `=`, `+=`, `-=`

Why do we need it?

1. To store or update values.
2. Makes code shorter.

How is it Used?

1. Used while declaring or changing variables.
2. Example: `x += 5`

6 Type Conversion

What is it?

1. Type conversion changes one data type into another.
2. Example: `int()`, `float()`, `str()`.

Why do we need it?

1. To work with different data types.
2. To avoid type errors.

How is it Used?

1. Used before calculations or input processing.
2. Example: `age = int(age)`

7 Input and Output

7.1 input()

What is it?

1. `input()` takes data from the user.
2. It always returns a string.

Why do we need it?

1. To get user information.
2. Makes programs interactive.

How is it Used?

1. Used to accept keyboard input.
2. Example: `name = input("Enter name: ")`

7.2 print()

What is it?

1. `print()` displays output on the screen.
2. It can print text and variables.

Why do we need it?

1. To show results.
2. Helps in testing programs.

How is it Used?

1. Used to display messages and outputs.
2. Example: `print(name)`

8 Strings

8.1 String Creation

What is it?

1. A string is created using single or double quotes.
2. Example: `"Python"`.

Why do we need it?

1. To store text.
2. Used for names and messages.

How is it Used?

1. Used wherever text is needed.
2. Example: `language = "Python"`

8.2 Indexing

What is it?

1. Indexing accesses one character from a string.
2. Index starts from 0.

Why do we need it?

1. To get a specific character.
2. Useful for text processing.

How is it Used?

1. Used to access individual characters.
2. Example: `text[0]`

8.3 Slicing

What is it?

1. Slicing gets part of a string.
2. Uses `start:end`.

Why do we need it?

1. To extract required text.
2. Makes string handling easier.

How is it Used?

1. Used to extract a substring.
2. Example: `text[0:5]`

8.4 String Operations

What is it?

1. Operations modify or combine strings.
2. Examples: `+`, `*`, `upper()`, `lower()`.

Why do we need it?

1. To process text.
2. To format output.

How is it Used?

1. Used to build or change text.
2. Example: `"Hello " + name`

9 Lists

9.1 Creating Lists

What is it?

1. A list stores multiple values.
2. It is created using square brackets `[]`.

Why do we need it?

1. To store many items together.
2. Makes data easy to manage.

How is it Used?

1. Used to group related values.
2. Example: `numbers = [1,2,3]`

9.2 Indexing

What is it?

1. Indexing accesses one element in a list.
2. Index starts from 0.

Why do we need it?

1. To get a specific item.
2. Useful for data access.

How is it Used?

1. Used to access a single list element.
2. Example: `numbers[1]`

9.3 Slicing

What is it?

1. Slicing gets multiple elements from a list.
2. Uses `start:end`.

Why do we need it?

1. To copy or extract data.
2. Makes list handling easier.

How is it Used?

1. Used to get a sublist.
2. Example: `numbers[1:4]`

9.4 Updating Elements

What is it?

1. List elements can be changed.
2. Assign a new value using an index.

Why do we need it?

1. To update stored data.
2. Lists are mutable.

How is it Used?

1. Used to modify existing data.
2. Example: `numbers[0] = 100`

9.5 Nested Lists

What is it?

1. A nested list is a list inside another list.
2. It stores grouped data.

Why do we need it?

1. To organize related data.
2. Useful for tables and matrices.

How is it Used?

1. Used to represent grid-like data.
2. Example: `matrix = [[1,2],[3,4]]`

9.6 Basic List Methods

9.6.1 append()

What is it?

1. Adds an item to the end of the list.
2. Changes the original list.

Why do we need it?

1. To add new data.
2. Useful when the list grows.

How is it Used?

1. Used to insert an item at the end.
2. Example: `numbers.append(10)`

9.6.2 insert()

What is it?

1. Inserts an item at a specific position.
2. Existing elements move right.

Why do we need it?

1. To add an item at any index.
2. Maintains order.

How is it Used?

1. Used to place an item at a chosen index.
2. Example: `numbers.insert(1,20)`

9.6.3 remove()

What is it?

1. Removes a specific value from the list.
2. Deletes the first matching item.

Why do we need it?

1. To delete unwanted data.
2. Keeps the list updated.

How is it Used?

1. Used to remove a known value.
2. Example: `numbers.remove(20)`

9.6.4 pop()

What is it?

1. Removes an element by index.
2. Returns the removed element.

Why do we need it?

1. To delete and use an item.
2. Useful in stack operations.

How is it Used?

1. Used to remove and retrieve an item.
2. Example: `numbers.pop()`

9.6.5 sort()

What is it?

1. Sorts list elements in ascending order.
2. Changes the original list.

Why do we need it?

1. To arrange data.
2. Makes searching easier.

How is it Used?

1. Used to order list values.
2. Example: `numbers.sort()`

9.6.6 reverse()

What is it?

1. Reverses the order of elements.
2. Changes the original list.

Why do we need it?

1. To display data in reverse order.
2. Useful in many algorithms.

How is it Used?

1. Used to flip element order.
2. Example: `numbers.reverse()`

9.6.7 len()

What is it?

1. `len()` returns the number of elements.
2. It works with lists, strings, and other collections.

Why do we need it?

1. To know the size of data.
2. Useful in loops and conditions.

How is it Used?

1. Used to check the count of items.
2. Example: `len(numbers)`

Assignment No: 2

Date: Wednesday, 5 August 2026

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1 List Methods

1.1 append()

What is it?

1. Adds an element to the end of a list.
2. Changes the original list.

Why do we need it?

1. To add new items.
2. To increase the list size.

How is it Used?

1. Use `append(value)`.
2. Example: `numbers.append(10)`

1.2 insert()

What is it?

1. Adds an element at a specific position.
2. Shifts other elements to the right.

Why do we need it?

1. To insert data at any index.
2. To keep the required order.

How is it Used?

1. Use `insert(index, value)`.
2. Example: `numbers.insert(1, 20)`

1.3 remove()

What is it?

1. Removes a specified element.
2. Deletes the first matching value.

Why do we need it?

1. To remove unwanted data.
2. To update the list.

How is it Used?

1. Use `remove(value)`.
2. Example: `numbers.remove(20)`

1.4 pop()

What is it?

1. Removes an element by index.
2. Returns the removed element.

Why do we need it?

1. To delete an item.
2. To use the removed value later.

How is it Used?

1. Use `pop(index)` or `pop()`.
2. Example: `numbers.pop()`

1.5 sort()

What is it?

1. Arranges elements in ascending order.
2. Changes the original list.

Why do we need it?

1. To organize data.
2. To make searching easier.

How is it Used?

1. Use `sort()`.
2. Example: `numbers.sort()`

2 Set Methods

2.1 add()

What is it?

1. Adds one element to a set.
2. Duplicate values are ignored.

Why do we need it?

1. To insert unique data.
2. To update the set.

How is it Used?

1. Use `add(value)`.
2. Example: `fruits.add("Apple")`

2.2 remove()

What is it?

1. Removes a specific element.
2. Gives an error if the element is missing.

Why do we need it?

1. To delete unwanted items.
2. To keep the set updated.

How is it Used?

1. Use `remove(value)`.
2. Example: `fruits.remove("Apple")`

2.3 discard()

What is it?

1. Removes a specific element.
2. Does not give an error if the element is missing.

Why do we need it?

1. To safely remove items.
2. To avoid program errors.

How is it Used?

1. Use `discard(value)`.
2. Example: `fruits.discard("Apple")`

2.4 clear()

What is it?

1. Removes all elements from the set.
2. Makes the set empty.

Why do we need it?

1. To reset the set.
2. To remove old data.

How is it Used?

1. Use `clear()`.
2. Example: `fruits.clear()`

2.5 union()

What is it?

1. Combines two sets.
2. Removes duplicate elements.

Why do we need it?

1. To merge sets.
2. To get all unique values.

How is it Used?

1. Use `set1.union(set2)`.
2. Example: `a.union(b)`

3 Tuple Methods

3.1 count()

What is it?

1. Counts how many times a value appears.
2. Returns an integer.

Why do we need it?

1. To find repeated values.
2. To analyze data.

How is it Used?

1. Use `count(value)`.
2. Example: `t.count(10)`

3.2 index()

What is it?

1. Finds the position of a value.
2. Returns the first matching index.

Why do we need it?

1. To locate an element.
2. To access data quickly.

How is it Used?

1. Use `index(value)`.
2. Example: `t.index(20)`

3.3 len()

What is it?

1. Returns the number of elements.
2. Works with tuples.

Why do we need it?

1. To know the tuple size.
2. To use in loops.

How is it Used?

1. Use `len(tuple_name)`.
2. Example: `len(t)`

3.4 max()

What is it?

1. Returns the largest element.
2. Works with numbers.

Why do we need it?

1. To find the maximum value.
2. Useful in calculations.

How is it Used?

1. Use `max(tuple_name)`.
2. Example: `max(t)`

3.5 min()

What is it?

1. Returns the smallest element.
2. Works with numbers.

Why do we need it?

1. To find the minimum value.
2. Useful in comparisons.

How is it Used?

1. Use `min(tuple_name)`.
2. Example: `min(t)`

4 Dictionary Methods

4.1 keys()

What is it?

1. Returns all keys.
2. Creates a view of keys.

Why do we need it?

1. To access key names.
2. Useful in loops.

How is it Used?

1. Use `keys()`.
2. Example: `student.keys()`

4.2 values()

What is it?

1. Returns all values.
2. Creates a view of values.

Why do we need it?

1. To access stored values.
2. Useful for processing data.

How is it Used?

1. Use `values()`.
2. Example: `student.values()`

4.3 items()

What is it?

1. Returns key-value pairs.
2. Creates a view of dictionary items.

Why do we need it?

1. To access both keys and values.
2. Useful in loops.

How is it Used?

1. Use `items()`.
2. Example: `student.items()`

4.4 get()

What is it?

1. Returns the value of a key.
2. Does not give an error if the key is missing.

Why do we need it?

1. To safely access data.
2. To avoid program errors.

How is it Used?

1. Use `get(key)`.
2. Example: `student.get("Name")`

4.5 update()

What is it?

1. Adds or updates key-value pairs.
2. Changes the original dictionary.

Why do we need it?

1. To modify data.
2. To add new information.

How is it Used?

1. Use `update()`.
2. Example: `student.update({"Age":22})`

5 Comparison of Data Structures

DS	Order	Duplicate	Index	Mutable
List `[]`	Yes	Yes	Yes	Yes
Set `{}`	No	No	No	Yes
Tuple `()`	Yes	Yes	Yes	No
Dict `{}`	Yes	Value = Yes, Key = No	Keyed	Yes

Table 1: Comparison of Python Data Structures (List, Set, Tuple, Dictionary)